



OPEN Enhancing groundwater vulnerability assessment framework under anthropogenic stressors

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Aquifer vulnerability assessment (AVA) is a critical tool for the protection of groundwater-dependent ecosystems, especially in regions under persistent anthropogenic stress. Yet conventional index-based methods often rely on static parameters that inadequately represent dynamic surface–groundwater interactions (SGWI) and evolving land use pressures. This study presents an enhanced AVA framework that integrates physically based SGWI simulations from a coupled SWAT–MODFLOW model with modified DRASTIC indices (DRASTIC, DRASTICL, and DRASTICLW), normalized using a spatially resolved Water Quality Index (WQI). The WQI served as an empirical proxy for contamination, strengthening the link between predicted vulnerability and observed groundwater quality. Key hydrogeological parameters are refined using calibrated SWAT–MODFLOW outputs to improve the representation of recharge, land-surface dynamics, and subsurface responses. The results show that moderate- to high-vulnerability zones dominate the catchment, particularly in cultivated lands and shallow aquifers. Areas with very high vulnerability, though limited, coincide with major water resources and industrial zones, indicating a practical need for targeted management interventions. Model evaluation using ROC analysis indicates improved discrimination performance from DRASTIC to DRASTICLW formulations. The integrated framework provides a physically consistent and empirically constrained approach for regional-scale vulnerability assessment, with applicability dependent on hydrogeological conditions and data availability. Comparative evaluations highlight notable shifts in vulnerability patterns when land use and water quality variables are included, confirming the influence of anthropogenic stressors. By integrating physically based hydrological modeling with empirical water quality data, this approach provides a transferable decision-support framework. It contributes to Sustainable Development Goal 6 (SDG 6) and supports adaptive groundwater management in environmentally sensitive regions.

Groundwater resources are essential components of the hydrological cycle, providing drinking water, agricultural irrigation, and support for industrial activities^{1–3}. Increasing anthropogenic pressures and hydroclimatic variability threaten these systems, underscoring the need for spatially explicit tools capable of diagnosing vulnerability beyond intrinsic susceptibility^{4,5}. Because groundwater contamination is complex and persistent, reliable vulnerability assessment is critical for informing management decisions and conservation planning. In regions experiencing strong human and environmental pressures, inadequate management undermines groundwater quality, sustainability, and ecosystem health^{6,7}. Effective assessments must integrate both natural and human-induced contributors to groundwater degradation to enhance aquifer sustainability and guide protective strategies^{8,9}. Aquifer vulnerability assessment (AVA) is therefore central to identifying pollution hotspots and implementing appropriate protection measures, with the underlying premise that the physical environment may partially shield aquifers from contamination^{10–12}.

Yet growing pressures from climate change and human activities continue to jeopardize groundwater quality, and inadequate or static vulnerability assessment frameworks limit the ability to safeguard this resource^{6,13,14}. Protecting groundwater from contamination requires robust methodologies that integrate intrinsic aquifer properties with contaminant transport dynamics¹⁵. AVA is commonly defined as “the tendency or likelihood for

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contaminants to reach a specified position in the groundwater system after introduction at some location above the uppermost aquifer¹⁶. A variety of AVA techniques have been developed using hydrogeological data^{13,15}. Index-based approaches remain widely used because of their simplicity and modest data requirements. Among them, the DRASTIC model—based on Depth to water table, Recharge, Aquifer media, Soil media, Topography, Impact of the vadose zone, and Conductivity—is one of the most common methods¹⁷. Numerous adjustments to DRASTIC have been proposed to improve its predictive performance in specific settings, including the addition of land use information (DRASTICL)¹⁸. The original and modified DRASTIC formulations have notable limitations: they struggle to quantify recharge accurately and ignore dynamic surface–groundwater interactions (SGWI), and they often perform poorly across diverse hydrogeological contexts^{19,20}. Recharge estimation is particularly challenging and prone to significant errors¹⁷. Despite widespread use, improvements in input parameter estimation remain limited, highlighting the need for multidimensional approaches to improve model comparison reliability and support sustainable groundwater management²¹.

Process-based hydrological models offer an opportunity to address these limitations. Models such as the Soil and Water Assessment Tool (SWAT)²² and MODFLOW²³ have been incorporated into DRASTIC/L frameworks to improve recharge and groundwater representation. SWAT has been used to refine recharge estimation within DRASTICL, demonstrating improved vulnerability mapping and support for pollution mitigation strategies^{24,25}. However, its recharge calculation is better suited to shallow aquifers and does not fully account for groundwater evapotranspiration, leading to inaccuracies in deeper unconfined systems. MODFLOW has been used to enhance DRASTICL by simulating aquifer dynamics and contaminant transport^{9,26}, yet it relies on prescribed recharge values and thus cannot reliably characterize spatial recharge variations. Consequently, SWAT is limited in representing deep aquifers, while MODFLOW is constrained by externally supplied recharge estimates. Recent advances in integrated modeling offer a pathway forward. Coupled frameworks such as SWAT–MODFLOW²⁷ effectively capture SGWI, improving the representation of aquifer dynamics and contaminant movement^{28,29}. Calibrated parameters from such integrated models can substantially reduce input uncertainty and strengthen AVA. Nonetheless, challenges persist, particularly related to data availability, spatiotemporal variability, and methodological uncertainty. These limitations highlight the need for robust, transferable AVA methodologies that perform effectively across diverse hydrogeological and hydroclimatic settings.

Traditional DRASTIC models rely solely on physical parameters and are often validated using single contaminants such as nitrate or total dissolved solids (TDS)³⁰. This pollutant-specific validation restricts their ability to capture the multi-pollutant conditions typical of groundwater contamination³¹. To overcome these constraints, this study introduces an integrated AVA framework that couples an optimized SWAT–MODFLOW hydrological model with a Water Quality Index (WQI). The resulting DRASTICLW model extends the conventional DRASTIC and DRASTICL structures by incorporating water quality (W) as an adaptive weighting parameter. The WQI serves as an empirical proxy for cumulative contamination, integrating the effects of multiple pollutants and anthropogenic pressures³². It is thus treated as an outcome-based indicator rather than a primary stressor. WQI-based modulation allows intrinsic hydrogeological vulnerability to be translated into contamination potential grounded in observed water quality conditions, thereby connecting aquifer sensitivity with real-world degradation patterns³³. This enables DRASTICLW to capture bidirectional feedbacks between hydrogeology and cumulative contamination, producing more realistic spatial delineations of vulnerability and enhancing early-warning capabilities^{34,35}. Given escalating pressures from climate change and anthropogenic activities, robust and integrated approaches to AVA are urgently needed. Integrating contamination indicators directly into weighting, DRASTICLW shifts vulnerability mapping from a static index to a dynamic, evidence-driven diagnostic tool aligned with adaptive groundwater management paradigms³⁶. Detecting emerging contamination risks and guiding resilient management require frameworks that incorporate physical hydrological processes while minimizing methodological uncertainty²⁰.

Accordingly, this study develops a comprehensive SWAT–MODFLOW–DRASTICLW framework to enhance AVA under regional-scale SGWI. Previous groundwater vulnerability studies have predominantly relied on static index-based methods or incorporated hydrological modeling without empirical constraints from water quality observations. The present study advances this body of work by coupling physically based surface–groundwater simulations with an enhanced DRASTIC framework and empirically constraining vulnerability patterns using observed groundwater quality data. This multi-layered integration strengthens both process representation and predictive performance, thereby addressing key limitations identified in earlier approaches. The framework provides spatially refined and physically consistent vulnerability estimates, enabling evidence-based decision-making and adaptive risk management to support sustainable groundwater management and protection. In this study, intrinsic vulnerability is the susceptibility of an aquifer to contamination based on hydrogeological conditions, whereas groundwater quality indicators represent observed contamination outcomes resulting from cumulative pressures. The proposed framework integrates these complementary perspectives by retaining the physical structure of intrinsic vulnerability while empirically constraining it using groundwater quality information.

Methodology

Design framework

The Gilgel Gibe Catchment is situated in the Omo–Gibe River Basin of southwestern Ethiopia and is a critical part of the hydrological system in the region. The catchment features a rugged landscape comprising plateaus that experience a humid tropical climate with monomodal rainfall distribution and a mix of semi-arid and humid subtropical conditions. The catchment has numerous intermittent rivers, contributing to the spatial and temporal hydrologic variability vital for domestic, agricultural, and ecological needs. The dominant soil types and slope variations significantly influence runoff dynamics, infiltration potential, and aquifer recharge capacity³⁷. Geogenic factors shape baseline hydrochemistry and contribute to contamination threats driven by both natural

and anthropogenic factors, which have critical effects on water resource management³⁸. Water quality challenges in Jimma reflect infrastructural and operational limitations, underscoring the socio-environmental importance of groundwater and public health. Therefore, forward-looking strategies must integrate aquifer vulnerability mapping with aquifer systems, climate-informed vulnerability indices, and improved monitoring frameworks to ensure long-term water security and resilient ecosystems. AVA has evolved significantly since the development of the original DRASTIC index by¹⁷.

The Soil and Water Assessment Tool (SWAT)²² is a continuous watershed model. It employs a semi-distributed approach to simulate daily hydrological processes, environmental impacts, and the effects of management practices. Although the SWAT model is used for groundwater simulation, it struggles to capture spatial and temporal variations in groundwater flow, limiting its use in areas with frequent surface–groundwater interactions (SGWI). MODFLOW³⁹, on the other hand, relies on specific input conditions, such as groundwater recharge and evapotranspiration, and overlooks their interactions with surface water. The coupled model addresses these limitations by improving input representation. The integrated SWAT–MODFLOW model combines SWAT and MODFLOW to represent SGWI processes²⁷. Model coupling involves intersecting SWAT hydrological response units (HRUs) and sub-basins with MODFLOW grid cells to facilitate SGWI. Disaggregated HRUs (DHRUs) intersecting with MODFLOW grid cells convey recharge and remaining evapotranspiration (ET) from the SWAT HRUs to the MODFLOW grid cells. DHRUs provide location and surface water flow data to MODFLOW grid cells, which then provide calculated hydraulic heads and interactive flow rates to SWAT on a daily basis²⁷. The deep infiltration volume simulated by SWAT is assigned to the MODFLOW grid cells as recharge. Unlike approaches that focus on specific parameters to characterise SGWI, the SWAT–MODFLOW model optimises AVA by fully considering SGWI characteristics. A coupled SWAT–MODFLOW model²⁸ was used to account for the effects of SGWI and to accurately estimate ET rates that determine net aquifer recharge. The model physically simulates SGWI, from which spatially distributed recharge, groundwater heads, and flow conditions were derived to parameterize key DRASTIC inputs. These fluxes are then transferred to MODFLOW as spatially distributed recharge, enabling a physically consistent representation of SGWI. The detailed model setup and calibration for SWAT–MODFLOW are provided in²⁸. The coupled SWAT–MODFLOW model was executed in transient daily mode (1997–2014), integrating surface hydrological processes with unconfined nonlinear groundwater flow and calibrated against observed streamflow and groundwater levels. Key hydrological parameters governing surface runoff, recharge generation, and groundwater flow were calibrated to reproduce observed streamflow and groundwater dynamics. Model calibration reduced discrepancies between simulated and observed values, thereby ensuring a reliable representation of recharge dynamics. This integration ensures that vulnerability parameters reflect internally consistent hydrological processes rather than independent empirical estimates.

This method was designed as a standardized and transferable framework for estimating intrinsic vulnerability, relying solely on hydrogeological parameters. The optimized SWAT–MODFLOW model parameters²⁸ were layered with geospatial datasets, which formed the basis for developing and applying the SWAT–MODFLOW–DRASTICLW AVA model. The model operates in transient mode at a daily time step, with coupling achieved by transferring deep percolation from SWAT hydrological response units to MODFLOW grid cells as recharge. The data preparation for the DRASTIC factors required defining the rating of each layer to reflect its respective relative significance. Aquifer properties such as permeability and fracture density govern pollutant attenuation⁴⁰. Geological maps from Ethiopian Ministry of Water and Energy (MoWE) were used to develop raster layers in ArcGIS, with assigned ratings and weights to characterize aquifer media and support AVA. Aquifer vulnerability to pollution conditioning parameters used for this particular application are derived based on^{28,37,41}.

Input datasets

AVA methods require physically representative inputs, which govern pollutant transport and are influenced by hydrological and hydrogeological factors¹⁷. Hydrogeological, climatic, land use, and groundwater quality datasets were sourced from national monitoring agencies, with spatial and temporal resolutions harmonized to support basin-scale vulnerability assessment. All datasets were subjected to quality control, including consistency checks, outlier screening, and harmonization to a common spatial framework prior to analysis. The coupled SWAT–MODFLOW framework²⁸ was used to derive key hydrological inputs, including recharge, hydraulic conductivity, and depth to groundwater, which were subsequently processed in a GIS environment. The model operates in transient mode at a daily time step, with SWAT simulating land-surface processes and deep percolation, and MODFLOW–NWT simulating groundwater flow in an unconfined system. Coupling is achieved by transferring deep percolation from SWAT hydrological response units (HRUs) to MODFLOW grid cells as spatially distributed recharge, with stream–aquifer exchange enabling dynamic surface–groundwater interactions. Daily model outputs were aggregated into multi-year mean spatial layers to represent long-term hydrogeological conditions relevant to regional vulnerability assessment. Recharge was computed as the mean of daily deep percolation fluxes, while groundwater depth was derived from simulated hydraulic heads relative to the ground surface. The model domain was discretized into a finite-difference grid. The depth to groundwater (D), defined as the distance between the ground surface and the water table, controls contaminant transport and attenuation capacity within the vadose zone¹². In this study, D was derived from transient SWAT–MODFLOW simulations for 1997–2014 and expressed as the long-term mean of daily groundwater levels, providing a representative measure of persistent hydrogeological conditions consistent with regional-scale AVA practices. The depth to the water table thematic layer was categorised into six ranges based on the DRASTIC ratings, ranging from lowest to highest, to indicate the level of vulnerability. While groundwater levels exhibit temporal variability, the use of long-term averaged conditions is appropriate for assessing intrinsic vulnerability rather than short-term fluctuations. For the weighted analysis, an overall weight of four was assigned, as described by¹⁷. The calculated rating map indicated that the rating ranged from 1 to 9, with an overall weight of 5 for the weighted analysis. Subsequently, a rating map for each parameter was developed and converted into a raster

format, and weights were assigned to generate a thematic layer. Each factor was mapped and classified into significant ranges that impact groundwater pollution.

Understanding the processes driving the factors and spatiotemporal variations of recharge requires the validation of hydrogeological formulations in multi-scale applications. Unconfined aquifers, which recharge more readily, also exhibit higher contamination risk, underscoring the need for integrated SGWI assessment. Net Recharge (R) is the amount of water that infiltrates the land surface and percolates through the unsaturated zone^{17,35,40}. Recharge estimation in the original DRASTIC framework is imprecise and often infeasible^{17,17} argued that the recharge calculation in the original DRASTIC framework is not feasible and is characterized by imprecise estimations. Recharge is a fundamental indicator of aquifer sustainability^{28,42}, yet it remains inherently difficult to quantify. This difficulty arises from an incomplete understanding of its controlling processes and spatiotemporal variability, as well as the inherent limitations of empirical relationships. To improve recharge representation, this study applied an integrated SWAT–MODFLOW model²⁸ that captures SGWI and yields more reliable evapotranspiration and recharge estimates. The model was calibrated using observed groundwater levels, streamflow, and aquifer parameters to reduce simulation errors. Recharge was computed as deep percolation simulated at a daily time step in SWAT and transferred to MODFLOW as spatially distributed flux. The resulting annual recharge outputs were processed in ArcGIS to produce a raster recharge map, which was assigned weights and classified into five recharge categories for vulnerability analysis.

Aquifer medium (A) is a crucial environmental attribute that maintains pollutants that cannot be absorbed by soil. The rock type affects the penetrability of aquifers in the groundwater. Hence, an accurate and integrated assessment of surface and subsurface parameters is crucial for effectively quantifying hydrologic processes. A raster map was created by dividing the original aquifer medium lithology map using a hydrogeological dataset. Soil texture and physicochemical properties control hydrological processes by influencing infiltration, runoff, and contaminant transport⁴³. Soil media (S) represents the uppermost portion of the vadose zone and contains substantial biological activity that influences pollutant movement¹⁷. It plays a significant role in controlling the vertical transport of contaminants into the vadose zone. A soil map was analysed and imported into ArcGIS to generate a raster thematic layer. The major soil classes were categorised into six rating classes and weighted according to their infiltration capacity. The gradient and direction of flow in an unconfined aquifer can be inferred from the overall slope of the landscape⁴³. Topography (T) influences surface processes but has limited control on subsurface contaminant transport. It is quantified by the land-surface gradient and affects the percolation of pollutants. Topography layers were classified into five categories and evaluated following principles used in^{17,30}.

The vadose zone (I) lies above the water table and is either unsaturated or intermittently saturated¹⁷. It plays a fundamental role in regulating contaminant migration by controlling percolation rates and influencing pollutant residence time in the unsaturated layer⁴³. Longer residence times promote attenuation, whereas more permeable vadose materials facilitate faster contaminant transport. In this study, lithological bore-well logs and drilling depth point data were analysed to characterise vadose zone properties. Subsurface lithology is a key determinant of contamination vulnerability because variations in material texture and structure strongly affect pollutant movement. Borehole information was used to quantify vadose zone influence following Eq. (1), incorporating weighted values of vadose thickness and dominant lithological units. Spatial distribution maps were generated using Inverse Distance Weighting (IDW) interpolation, which was selected to balance local variability with spatial smoothness given the uneven sampling density. Sampling density is higher in the northern basin due to better accessibility and monitoring infrastructure; consequently, interpolation uncertainty increases in sparsely sampled areas, and the resulting maps should be interpreted with this limitation in mind. The resulting vadose zone indices were imported into ArcGIS and spatially interpolated using the IDW method to generate a continuous raster surface representing the impact of the vadose zone. These interpolated values were then classified into rating categories to assess the contribution of the vadose zone to contaminant transport toward groundwater.

Hydraulic conductivity (C) is a fundamental property that governs the rate at which groundwater migrates toward the saturated zone. Consequently, aquifer conductivity is directly linked to groundwater susceptibility. As described in¹⁷, hydraulic conductivity controls both the ability of an aquifer to transmit water and the speed at which groundwater moves. Higher conductivity increases the likelihood that contaminants are transported within the saturated zone. Optimized conductivity values were obtained from the integrated SWAT–MODFLOW model framework to provide a more reliable output. This reduced the input uncertainty for the AVA, thereby producing more reliable and accurate results. The conductivity raster layer was produced in ArcGIS to represent the magnitude and spatial weighting of conductivity-related factors. Land use (L) influences the movement of pollutant loads by altering nitrate levels in the unsaturated zone^{44,45}. Accordingly, L⁴¹ was incorporated as a DRASTIC modifier to improve the AVA. Land use categories were classified into six groups and integrated into the model, with weights, ranges, and ratings assigned according to the recharge capacity of each class following⁴⁵. The incorporation of depth and multi-layer lithological structure provides a quantitative refinement that accounts for heterogeneous vadose zones^{15,20,25,35,40,46}. The vadose zone parameter (Eq. (1)), was computed as a weighted average representing attenuation potential across the aquifer profile, from the ground surface to the base of the aquifer, following the concepts of^{17,40,47–49}.

$$I = \frac{1}{D_T} \sum_{i=1}^n D_{L_i} R_{L_i} \quad (1)$$

where I is the impact of the vadose zone; D_T is the depth to the water table from the Earth's surface; D_{L_i} is the depth (thickness) of lithological unit i ; and R_{L_i} is the rating assigned to lithological unit i based on its material properties.

Vulnerability index estimation

The final modified DRASTICLW vulnerability model was developed by incorporating L and WQI into the original model as anthropogenic stressors. Geographic information systems (GIS) were used to generate thematic layers. DRASTIC/L considers hydrological and geological factors that affect the movement of pollutants into the groundwater. Ranges, ratings, and weights were assigned to each hydrogeological layer to determine the influence of the various criteria and evaluate their contributions to the AVA. The AVA method assigns a weight to each parameter, which falls within the ranges from 1 to 5. Rating values ranging from 1 to 10 indicate ratings corresponding to discrete conditions based on¹⁷. The rating for each input layer was multiplied by the weight of each layer and summed to determine the vulnerability index (VI). Both the DRASTIC Eq. (2) and DRASTICL Eq. (3) vulnerability indices calculated as the sum of the products of the weight and rating for each parameter and produced using a weighted sum overlay analysis in the ArcGIS environment. The rates and weights of each raster pixel were accurately assigned based on rating maps and comprehensive workflow diagram of the methodology is as shown in (Fig. 1).

$$V_{\text{DRASTIC}} = (D W_D) + (R W_R) + (A W_A) + (S W_S) + (T W_T) + (I W_I) + (C W_C) \quad (2)$$

$$V_{\text{DRASTICL}} = (D W_D) + (R W_R) + (A W_A) + (S W_S) + (T W_T) + (I W_I) + (C W_C) + (L W_L) \quad (3)$$

where: D = Depth to water table, R = Net recharge, A = Aquifer media, S = Soil media, T = Topography, I = Impact of the vadose zone, C = Hydraulic conductivity, L = Land use / land cover parameter, and $W_D, W_R, W_A, W_S, W_T, W_I, W_C, W_L$ are the corresponding weights.

Vulnerability index classification

The continuous DRASTIC vulnerability index (VI) was transformed into discrete vulnerability zones for hydrogeologic interpretation and management prioritization. This classification was based on the mean (μ) and standard deviation (σ) of the VI distribution. This approach provides a reproducible, data-driven partitioning, while retaining inherent subjectivity associated with threshold selection. The composite DRASTIC index was generated through the weighted linear combination of standardized parameters. The resulting raster represents the intrinsic aquifer vulnerability field. The spatial distribution of VI values was partitioned into five statistically defined vulnerability classes using $\mu \pm k\sigma$ thresholds^{30,40}. This method assumes a quasi-normal distribution of VI across the study area and provides reproducible classification boundaries derived from the underlying data structure. This framework ensures that areas exceeding $\mu + 1.5\sigma$ correspond to the upper tail of the VI distribution and indicate elevated aquifer susceptibility. In contrast, areas below $\mu - 1.5\sigma$ represent hydrogeologically protected regions with strong natural barriers to contaminant migration. By deriving thresholds directly from statistical descriptors, the classification maintains consistency across vulnerability scenarios (Table 1), thereby supporting comparative vulnerability assessment under projected hydroclimatic and anthropogenic changes.

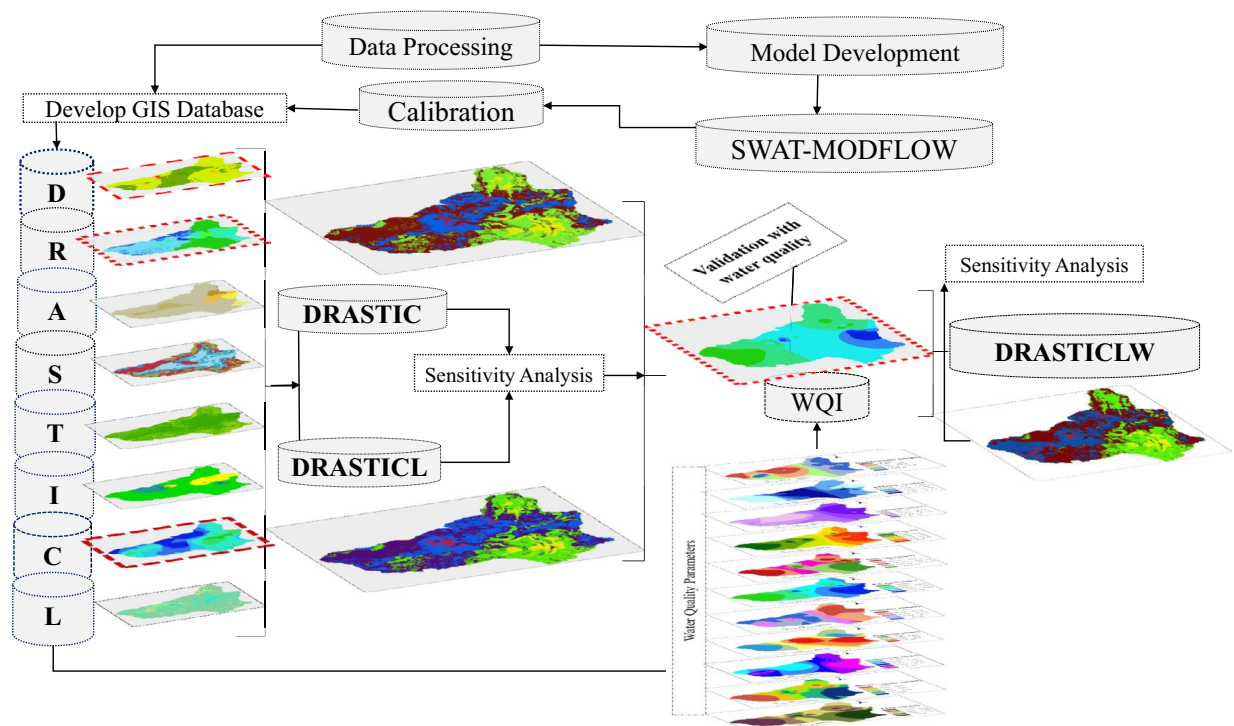


Fig. 1. Workflow diagram of the study methodology including aquifer and water quality controlling parameters for AVA.

The μ - σ -based classification provides a rigorous and reproducible means of delineating vulnerability zones. The classification scheme is designed for consistent spatial interpretation and robustness analysis rather than strict statistical inference. Regions in the upper quantiles ($VI > \mu + 1.5\sigma$) represent critical zones requiring priority management. In contrast, regions in the lower quantiles ($VI < \mu - 1.5\sigma$) represent relatively secure hydrogeologic domains. This framework enhances interpretability, reproducibility, and comparability of aquifer vulnerability maps across spatial scales. The μ and σ of the VI were computed as Eq. (4) and Eq. (5).

$$\mu = \frac{1}{n} \sum_{i=1}^n VI_i \tag{4}$$

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (VI_i - \mu)^2} \tag{5}$$

where VI_i denotes the vulnerability index value at grid cell i , and n is the total number of valid cells.

The μ and σ provide a statistically robust framework for classifying DRASTIC vulnerability maps by summarizing the intrinsic variability of the composite index and translating continuous spatial data into deviation-based risk zones. The mean represents the baseline hydrogeologic state, while the standard deviation quantifies spatial heterogeneity around this baseline⁵⁰. In heterogeneous aquifer systems, where lithologic, recharge, and soil properties vary continuously, the μ - σ method offers an objective means to distinguish statistically normal from extreme vulnerability zones^{30,40}. Spatially, values within $\mu \pm 0.5\sigma$ denote moderate vulnerability–hydrogeologic equilibrium with balanced recharge and filtration. Zones above $\mu + 1.5\sigma$ indicate very high vulnerability, corresponding to shallow water tables, coarse lithology, or intense recharge that accelerate contaminant transport. Values below $\mu - 1.5\sigma$ mark very low vulnerability, typifying confined or fine-textured formations providing strong natural protection¹⁷. Deriving class thresholds from $\mu \pm k\sigma$ ensures objectivity and reproducibility, avoiding arbitrary or visually optimized breaks, and allowing consistent temporal and spatial comparisons³⁵. Unlike other methods, this approach (Table 1) captures the statistical tails of the DRASTIC distribution, improving sensitivity to extreme conditions while preserving interpretability⁵¹. Each deviation from μ quantitatively reflects how unusual a vulnerability condition is, embedding a probabilistic interpretation that aligns with uncertainty quantification frameworks. Consequently, μ - σ classification maintains statistical transparency and hydrologic relevance, providing a strong basis for identifying high-risk aquifer zones and supporting risk-informed groundwater management. Thresholds of $\mu \pm 0.5\sigma$ and $\mu \pm 1.5\sigma$ were used to define five vulnerability classes relative to the basin-wide mean. This scheme aligns with common AVA practice, balancing interpretability with sufficient resolution to delineate priority zones.

Parameter sensitivity analysis

AVA is indispensable for understanding aquifer vulnerability to contamination and guiding sustainable watershed management. Analyzing key parameters uncovers spatial variations that are essential for comprehending the intricacies of vulnerability. Properly assessed prediction uncertainty is vital for effective decision-making, optimizing data acquisition, and resource protection⁵². Map removal SA is a GIS-based spatial analysis method that assesses how excluding certain controlling features impacts the analysis outcomes, yielding pivotal insights into the dynamics of vulnerability. The extensive use of input data based on GIS in DRASTIC, DRASTICL, and DRASTICLW reduces the errors.⁵³ demonstrated that incorporating land use data and conducting sensitivity-based parameter adjustments in the DRASTIC model significantly improved agreement with observed water quality metrics. The SA method systematically assesses the impact of removing individual parameters on overall vulnerability. However, this is challenging because of the variety of input data and the subjective parameter selection. Sensitivity analysis (SA) is employed to handle the subjectivity and variability in the model parameters by assessing how changes in the parameter values affect the overall results. A map-removal sensitivity analysis was used to evaluate the impact of removing one or more layers⁵⁴. Subsequently, the mutual reliance and variability of the layers were evaluated to minimize erroneous judgments. This analysis involved removing one parameter at a time and measuring its effect on the overall vulnerability score of the remaining parameters. The VI obtained using both DRASTIC and DRASTICL parameters was regarded as unperturbed, whereas the index calculated using a reduced number of data layers was considered perturbed³⁰. The map removal sensitivity measure defined by⁵⁴ is expressed as shown in Eq. (6).

Class	Statistical Range	Interpretation
Very Low	$VI \leq (\mu - 1.5\sigma)$	Strong protective cover; low-permeability
Low	$(\mu - 1.5\sigma) < VI \leq (\mu - 0.5\sigma)$	Low recharge; thick unsaturated zone
Moderate	$(\mu - 0.5\sigma) < VI \leq (\mu + 0.5\sigma)$	Balanced attenuation potential
High	$(\mu + 0.5\sigma) < VI \leq (\mu + 1.5\sigma)$	High permeability
Very High	$VI > (\mu + 1.5\sigma)$	Critical zones with minimal protection

Table 1. Statistical classification scheme for vulnerability.

$$S = \left(\frac{\frac{V}{N} - \frac{V'}{n}}{V} \right) \times 100 \quad (6)$$

where S represents the sensitivity index, V denotes the unperturbed vulnerability index, V' signifies the perturbed VI after map removal, and N and n represent the number of data layers used to compute V and V' , respectively.

Water quality assessment

Water quality parameters were characterized according to the criteria provided by the World Health Organization (WHO). Groundwater quality parameters were derived from monitoring data collected across multiple campaigns and aggregated to represent conditions relevant to vulnerability assessment rather than short-term variability. Data were processed to annual mean concentrations at each sampling location; where multiple observations existed within a year, values were averaged prior to spatial interpolation and used to construct the WQI map. The resulting surface represents long-term groundwater quality and integrated contamination consistent with the modeled hydrogeological regime. This approach supports identification of contamination-prone areas rather than analysis of temporal trends. The pH level indicates the acidity or alkalinity of a specific soil solution by measuring the concentration of hydrogen ions on a logarithmic scale ranging from 0 to 14, controlling geochemical equilibrium. The pH level ranged from 5.83 to 8.17, with a mean of 7.17 and a standard deviation (SD) of 0.31. According to^{55,56}, the pH range suitable for drinking water is between 6.5 and 8.5, and all samples fell within the permitted limits, with a slightly alkaline nature (Fig. 2a). Groundwater infiltration and the decomposition of chemical substances from agricultural runoff contribute to high levels of electrical conductivity (EC), which measures water's ability to conduct electricity. The EC varied from 162.67 to 1544.98 μ S/cm, with a mean of 693.91 μ S/cm and SD of 119.03 μ S/cm, which is within permitted limits (Fig. 2b). The total dissolved solids (TDS) measure the salinity of groundwater, revealing the types of minerals present in the water. The TDS ranged between 152 and 979.99 mg/L, with a mean of 128.3 mg/L and SD of 119.29 mg/L. TDS indicates the quantity of salts dissolved in water, which often originate from natural sources through sediment leaching. High TDS values may result from prolonged groundwater residence times. Only the spatial patterns over the watershed are shown in (Fig. 2c) as no specific health-based threshold exists for TDS in drinking water. Calcium (Ca^{2+}) originates from both natural and anthropogenic sources, including aquifer dissolution of naturally occurring rocks and minerals. The Ca^{2+} concentration ranged from 4.46 to 72.09 mg/L, with a mean of 23.40 mg/L and SD of 5.0 mg/L.

According to the WHO, the maximum permissible level of Ca^{2+} is 75 mg/L, and all samples met this standard⁵⁶ (Fig. 2d). Sodium (Na^+), a highly reactive alkali metal in groundwater, varied from 25.5 to 389.99 mg/L, with a mean of 128.3 mg/L and SD of 51.29 mg/L. Concentrations exceeding 200 mg/L are considered unsuitable for human consumption (Fig. 2e). Potassium (K^+), present in various minerals and rocks, often results from wastewater discharge and fertilizer use. The K^+ concentration ranged from 1.79 to 24.78 mg/L, with a mean of 8.29 mg/L and SD of 0.92 mg/L (Fig. 2f). Magnesium (Mg^{2+}) influences water hardness. The measured Mg^{2+} values ranged from 1.46 to 27.61 mg/L, with a mean of 7.42 mg/L and SD of 1.14 mg/L, all within the WHO permissible limit of 30 mg/L (Fig. 2g). Chlorides (Cl^-), originating from weathering, rocks, soils, and wastewater discharge, ranged from 1.01 to 39.96 mg/L, with a mean of 7.38 mg/L and SD of 1.49 mg/L (Fig. 2h). Fluoride (F^-), naturally present in groundwater, ranged from 0.82 to 10.99 mg/L, with a mean of 4.42 mg/L and SD of 0.95 mg/L, within acceptable limits for drinking water (Fig. 2i). Nitrate (NO_3^-), an important ion in the nitrogen cycle, varied from 0.08 to 1.96 mg/L, with a mean of 0.50 mg/L and SD of 0.14 mg/L (Fig. 2j). Sulfate (SO_4^{2-}) concentrations, essential for assessing groundwater quality, ranged from 0.12 to 119.39 mg/L, with a mean of 8.54 mg/L and SD of 4.0 mg/L (Fig. 2k). Bicarbonate (HCO_3^-), produced by soil organic matter breakdown and CO_2 dissolution in precipitation, ranged from 131.76 to 995.5 mg/L, with a mean of 448.56 mg/L and SD of 161.47 mg/L. All samples were within the permissible WHO limits⁵⁶ (Fig. 2l). The process of spatial mapping of water quality measures was carried out to understand the variations, assist in detecting polluted areas, and prepare mitigation strategies.

Model normalization

Water management is complicated by human activities that pollute water resources. Groundwater has been overused because of the need to quickly consume resources, causing pollution, water stress, and decreased availability and accessibility. The Water Quality Index (WQI) is essential for groundwater evaluation and management⁴⁸. The WQI is a globally accepted tool for measuring human impacts on groundwater quality and informing decision-makers of environmental conditions⁵⁷. Groundwater physicochemical parameters were weighted based on their importance and combined into a single index to better assess contamination levels⁵⁸. A weighted arithmetic approach was used to compute the WQI of the groundwater samples. Weighted index approaches analyze water quality by weighting each characteristic based on its importance. Weights assigned to individual water quality parameters follow established WQI formulations and reflect their relative significance for potable groundwater assessment, consistent with WHO guidelines⁵⁶. Geospatial maps of the WQI were used to validate both DRASTIC and DRASTICL models. The spatial distributions of regional water quality controlling Parameters over the watershed are as indicated in (Fig. 2l). The relative weight (W_i) was calculated using the following equation Eq. (7). Each sample parameter was assigned a water quality rating scale (q_i), which was determined using Eq. (8). Then, using Eq. (9), the overall groundwater WQI parameters were evaluated. A central contribution of this work lies in the integration of groundwater quality information through a normalized WQI. Its incorporation as a scaling modifier allows the intrinsic vulnerability structure derived from hydrogeological conditions to be adaptively constrained by observed water quality patterns.

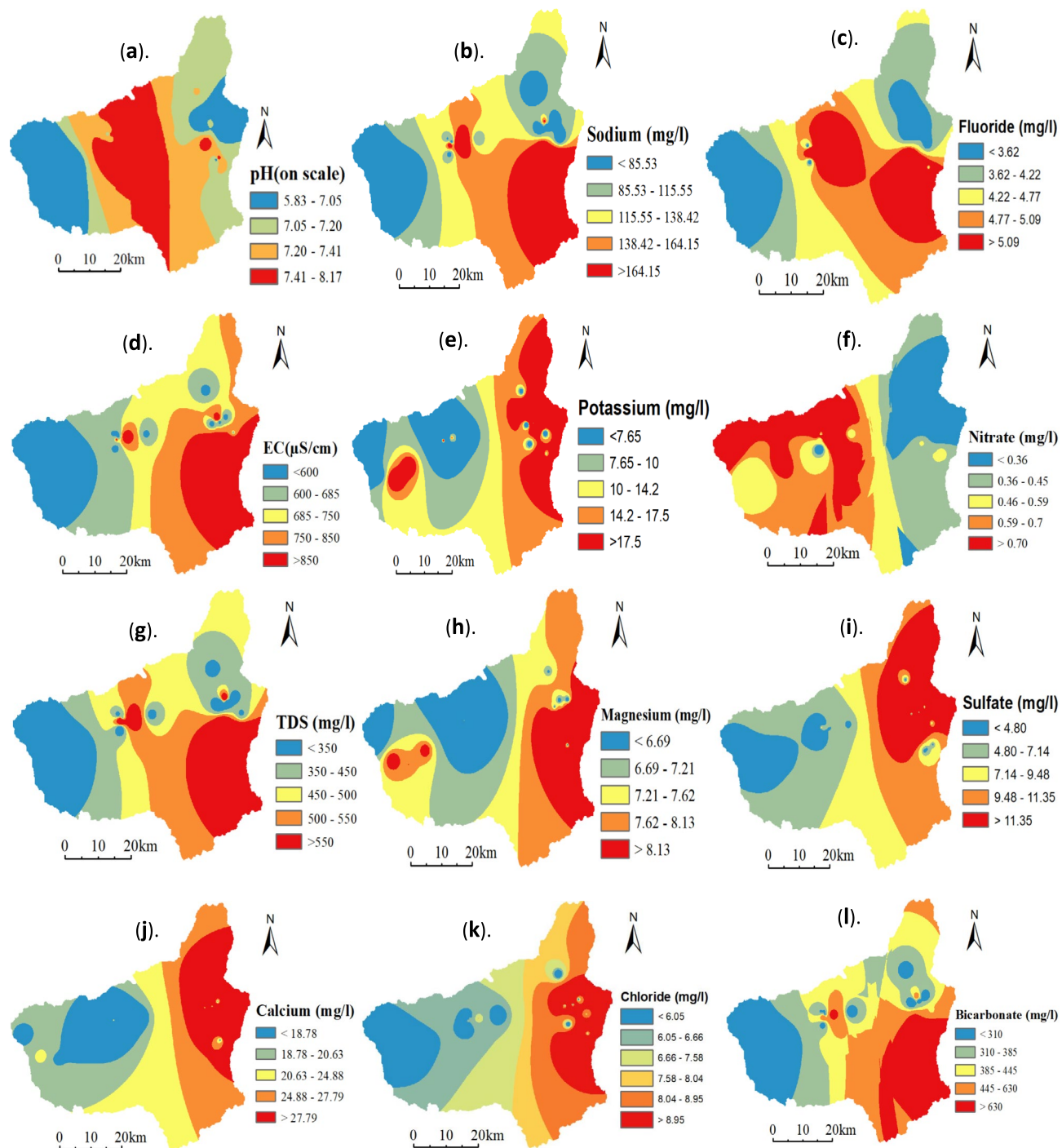


Fig. 2. Spatial distribution of regional water quality controlling parameters: (a). pH; (b). EC; (c). TDS; (d). Ca^{2+} ; (e). Na^{+} ; (f). K^{+} ; (g). Mg^{2+} ; (h). Cl^{-} ; (i). F^{-} ; (j). NO_3^{-} ; (k). SO_4^{2-} ; (l). HCO_3^{-} .

$$W_i = \frac{w_i}{\sum_{i=1}^n w_i} \quad (7)$$

$$q_i = \frac{C_i}{S_i} \times 100 \quad (8)$$

$$\text{WQI} = \sum_{i=1}^N W_i q_i \quad (9)$$

where W_i is the relative weight, w_i is the weight of each parameter, and n is the total number of parameters; q_i is the quality rating, C_i is the concentration of each physicochemical pollutant in a water sample (in mg L^{-1}), and S_i is the corresponding WHO standard concentration; and WQI is the weighted water quality–refined DI predictor.

Developing comprehensive vulnerability index

Given the widespread adoption of various AVA methods, identifying their potential shortcomings is essential to avoid over- or underestimating the contamination risk delineation. This is critical for refining the AVA, prioritizing management actions, and informing sustainable groundwater management strategies. Effective aquifer management requires a comprehensive understanding of aquifer characteristics, contaminant sources, and hydrological processes to mitigate pollution risks. Incorporating WQI into AVA refines the method by considering water quality alongside other factors and is designated as DRASTICLW, where W represents the WQI (Fig. 1). This study advances AVA by integrating empirical groundwater quality information via a normalized WQI as a proxy for reflecting cumulative contamination outcomes, thereby providing an independent and quantitative basis for constraining and evaluating model-derived vulnerability patterns. This new vulnerability category is defined based on specific ranges that supplement the study outcomes of DRASTIC and DRASTICL AVA. The inclusion of the WQI as a controlling parameter indicates that the AVA framework accounts for the water quality determinants. Therefore, a more comprehensive and enhanced vulnerability assessment framework (DRASTICLW) was developed. The mathematical formulation of DRASTICLW is given in Eq. (11), where the integration of normalized WQI (Eq. (10)), within the DRASTIC framework constitutes the core methodological advancement of the proposed DRASTICLW model. The WQI functions as a weight-modulation factor, a dynamic scaling coefficient that continuously adjusts the relative influence of hydrogeological parameters according to the empirical contamination status of the aquifer. This mechanism transforms the traditional, expert-assigned weighting scheme of DRASTIC into a data-adaptive and contamination-responsive system, effectively bridging intrinsic aquifer susceptibility with observed groundwater quality deterioration. To introduce adaptability, the composite WQI was computed from multiple physicochemical indicators Eq. (10). This normalization ensures inter-regional comparability and prevents extreme WQI values from disproportionately influencing the composite index. The normalized WQI is subsequently integrated into the DRASTIC framework as a modulation factor, producing the composite VI as in Eq. (11). The DRASTICLW model improves robustness by coupling intrinsic hydrogeologic factors with empirical water quality metrics, reduces subjectivity in weight assignment, enhances generalization through the use of normalized (WQI_{norm}), and maintains scalability while preserving the conceptual integrity of the original DRASTIC framework.

$$WQI_{\text{norm}} = \frac{WQI - WQI_{\min}}{WQI_{\max} - WQI_{\min}} \quad (10)$$

$$V_{\text{LW}} = \left(\sum_{i=1}^8 R_i W_i \right) \times WQI_{\text{norm}} \quad (11)$$

where, V_{LW} represents the DRASTICLW VI, wherein WQI_{norm} amplifies or attenuates intrinsic DRASTIC and DRASTICL score in proportion to the degree of groundwater quality degradation.

Eq. (11) incorporates normalized WQI as a scaling modifier that empirically constrains intrinsic vulnerability patterns based on observed groundwater quality, while preserving the spatial structure derived from hydrogeological conditions. Higher WQI_{norm} values indicate poorer water quality and therefore intensify the index, highlighting hotspots where hydrogeologic weakness aligns with chemical degradation, while lower WQI_{norm} values suppress scores, preventing overestimation and ensuring a balanced representation of AVA. Min–max normalization was applied to scale WQI values to ensure compatibility with the intrinsic VI and to preserve relative proportions across the basin. Acting as a proxy for contamination intensity, the WQI_{norm} enables adaptive reweighting of parameters according to prevailing hydrochemical conditions. This feedback-driven modulation imparts adaptive mechanisms to the model, allowing the vulnerability surface to evolve dynamically with spatiotemporal variations in groundwater quality while reducing underfitting by detecting overlooked contamination, and overfitting by moderating exaggerated vulnerability in chemically stable zones. Consequently, DRASTICLW transforms DRASTIC/L from a static structural model into a hybrid intrinsic-specific diagnostic, capable of representing cumulative anthropogenic and geochemical stress. The multiplicative formulation emphasizes interaction effects among parameters and reduces compensatory behavior inherent in additive aggregation, thereby improving differentiation of vulnerability patterns⁵⁹. Multiplicative integration is not intended as a purely empirical adjustment but reflects the conceptual interaction between intrinsic hydrogeological susceptibility and observed contamination conditions. Thus, the normalized WQI serves as a modulation factor that scales intrinsic vulnerability based on empirical groundwater quality conditions.

The receiver operating characteristic (ROC) curve is widely used to evaluate and compare predictive model performance^{40,43}. ROC analysis evaluates the ability of the model to discriminate spatial gradients of contamination susceptibility rather than binary classification outcomes. Model outcomes are categorized as true positives (high vulnerability predictions correctly coinciding with observed contamination), false positives (high vulnerability predicted where contamination is absent), true negatives (low vulnerability predictions in uncontaminated areas), and false negatives (low vulnerability predicted despite contamination being present). From these outcomes, the true positive rate (sensitivity) and false positive rate are computed, illustrating the trade-off between successfully detecting contamination and mistakenly flagging uncontaminated areas as high-risk⁴⁹. The area under the ROC curve (AUC–ROC) provides a threshold-independent measure of overall

discriminatory power, indicating how effectively a model distinguishes contaminated from uncontaminated areas^{60,61}. AUC–ROC has been applied to compare the performance of different AVA methods^{43,62}. Hence, the ROC curve was used to evaluate the performance of aquifer vulnerability zoning tasks on each AVA model to evaluate spatial patterns of contamination susceptibility rather than discrete contamination states. This metric improves evaluation transparency and provides a robust basis for risk-informed decision-making. It delivers a more realistic, data-driven, and transferable measure of groundwater risk, uniting hydrogeologic predisposition and empirical contamination evidence into a coherent, policy-relevant framework for adaptive water-resource management.

Results and discussion
Aquifer parameter analysis

AVA assumes that groundwater contamination is initiated at the surface and is then transported through the unsaturated zone to the saturated zone²¹. Parameters with a greater likelihood of controlling contaminant transport were assigned numerical values. The sub-classes and rating of depth (m): < 4.77 (1), 4.77–13.44 (2), 13.44–20.14 (3), 20.14–28.89 (5), 28.89–35.29 (7), and > 35.29 (9) cover 0.26%, 0.33%, 2.61%, 63.33%, 31.64%, and 1.83% respectively of the total area. The dominant water table depths fell within classes 5 and 7. The lowest rate underscores areas where the water table is shallow, thereby increasing the risk of contamination. Conversely, a high maximum rate in the aquifer media indicates the presence of highly permeable materials, facilitating rapid contaminant transport through the aquifer. Shallower depths are more prone to contamination and, consequently, are given a higher ranking. Areas with deeper groundwater levels were assigned lower ratings because contaminants require a longer travel path before reaching the aquifer. The relationship between this parameter and the total vulnerability assessment indicates that a higher water table depth corresponds to a decreased likelihood of pollution, and vice versa¹⁷.

The optimized coupled SWAT–MODFLOW model²⁸ demonstrated strong hydrological consistency (Nash–Sutcliffe efficiency (NSE = 0.87), coefficient of determination ($R^2 = 0.84$), and percent bias (PBIAS = −8.73%), indicating that the model adequately represents watershed-scale water balance and recharge dynamics. These results support its use for generating spatially distributed hydrogeological inputs, while not constituting full validation of subsurface flow and transport processes. The coupled SWAT–MODFLOW simulations reveal spatially heterogeneous recharge patterns and groundwater depths across the basin, reflecting the combined effects of topography, land use, and hydrogeological conditions. Shallow groundwater depths prevail in lowland agricultural areas, thereby increasing vulnerability to surface-derived contaminants. These physically based simulations directly inform the spatial structure of groundwater vulnerability indices. Accordingly, net recharge (mm) is classified as with a rating of: < 45.13 (1), 45.13–90.25 (3), 90.25–135.38 (5), 135.38–180.5 (7), 180.50–225.63 (9), and > 225.63 (10) covering 33.81%, 4.15%, 19.03%, 31.54%, 9.46%, and 2.01% respectively of the total drainage area. The current analysis surpasses previous studies on AVA by incorporating a more realistic recharge parameter rather than relying solely on traditional average precipitation calculations.

The aquifer media were classified into six hydrogeological units: upper felsic volcanic, alluvial–gravel, sandy pyroclastic, sedimentary rock, lower felsic volcanic, and volcanic sand. Each unit was assigned a rating on a scale following the DRASTIC framework, based on its relative permeability and attenuation capacity as reported in previous studies¹⁷. These units occupy 77.1%, 8.06%, 5.47%, 2.99%, 3.51%, and 2.88% of the study area, respectively. A uniform weight of 3 was applied to the aquifer media parameter in accordance with the standard weighting scheme (Table 2). Soil media (S), the biologically active upper vadose zone, significantly influences pollutant migration¹⁷. Soil (S) properties control infiltration and contaminant transport through the unsaturated zone. These classifications were based on permeability, adsorption capacity, and attenuation behavior^{17,30,40}. Spatial analysis indicates that Dystric Nitisols (32.98%) and Eutric Fluvisols (24.04%) dominate the study area, followed by Dystric Fluvisols (15.11%) and Chromic Vertisols (10.22%), whereas Orthic Acrisols (7.24%) and Eutric Nitisols (0.41%) are less extensive (Table 2). The prevalence of Nitisols and Fluvisols implies predominantly moderate-to-high permeability conditions, likely enhancing vertical recharge and increasing groundwater vulnerability, whereas the limited extent of Acrisols and Vertisols suggests a comparatively lower contribution to vulnerability. Aquifer media (A) represents the consolidated or unconsolidated geological formations that store and transmit groundwater^{17,30}.

Parameter	Range	Rating (1–10)	Weight
D	<4.77 / 4.77–13.44 / 13.44–20.14 / 20.14–28.89 / 28.89–35.29 / >35.29	1 / 2 / 3 / 5 / 7 / 9	5
R	<45.13 / 45.13–90.25 / 90.25–135.38 / 135.38–180.50 / 180.50–225.63 / >225.63	1 / 3 / 5 / 7 / 9 / 10	4
A	Felsic volcanic / Gravel alluvium / Pyroclastic / Sedimentary / Volcanics / Volcanic sand	2 / 5 / 6 / 7 / 8 / 10	3
S	Chromic Vertisols / Dystric Fluvisols / Dystric Nitisols / Eutric Fluvisols / Orthic Acrisols / Eutric Nitisols	2 / 4 / 6 / 8 / 9 / 10	2
T	<11.03 / 11.03–22.06 / 22.06–33.09 / 33.09–44.12 / 44.12–55.15 / >55.15	9 / 8 / 7 / 5 / 3 / 1	1
I	<4.98 / 4.98–5.64 / 5.64–6.31 / 6.31–6.97 / 6.97–7.64 / >7.64	1 / 2 / 3 / 5 / 9 / 10	4
C	<10.5 / 10.5–20.5 / 20.5–41.76 / 41.76–49.99 / 49.99–62.16 / >62.16	3 / 5 / 6 / 7 / 9 / 10	3
L	Bare / Built-up / Agricultural / Grassland / Forest / Waterbody	2 / 3 / 9 / 5 / 4 / 1	2

Table 2. Classification ranges, rating, and assigned weights for controlling parameters following¹⁷, extended with land use.

AVA is strongly influenced by the unsaturated zone, which controls contaminant infiltration and is shaped by the properties of the underlying aquifer media. The unsaturated zone is situated between the soil layer and water table. Topography was classified as <11.03 (1.98%), 11.03–22.06 (6.56%), 22.06–33.09 (14.17%), 33.09–44.12 (23.03%), 44.12–55.15 (29.46%), and >55.15 (24.76%)³⁷. The vadose zone classified as: <4.98 (1), 4.98–5.64 (2), 5.64–6.31 (3), 6.31–6.97 (5), 6.97–7.64 (9), and > 7.64 (10) with an overall areal coverage of 0.28%, 10.2%, 75.67%, 13.34%, 0.44%, and 0.07% respectively. A weight of 4 was assigned (Table 2), as a thicker vadose zone slows percolation into the aquifer⁶³. Understanding aquifer properties is essential for vulnerability assessments because they directly affect the speed at which contaminants can move into the aquifer. Areas with higher weighted values in the vadose zone were assigned higher ratings, indicating greater vulnerability to pollution, and vice versa. The conductivity (m) was classified as < 10.5, 10.5–20.5, 20.5–41.76, 41.76–49.99, 49.99–62.16, and > 62.16 with a rating and areal coverage of 3 (12.02%), 5 (22.33%), 6 (32.28%), 7 (20.48%), 9 (11.79%), and 10 (1.11%), respectively, followed by three overall weights (Table 2). Low-lying areas are more susceptible to surface water infiltration and contamination from agricultural runoff, industrial discharges, and urban effluents, whereas steeper slopes generally produce more runoff and reduce aquifer vulnerability.

Topography was assigned the lowest weight (1) (Table 2), because slope mainly regulates the partitioning between surface runoff and infiltration at the land surface, but does not influence contaminant transport once water infiltrates beyond the soil and vadose zone. Slope-induced effects dissipate below the unsaturated profile, and therefore topography exerts no direct control on pollutant migration within the aquifer. Topography exhibits relatively low sensitivity, consistent with its minimal assigned weight, indicating a limited influence on vulnerability outcomes at the basin scale. Sensitivity analyses consistently demonstrate that topography contributes the least to the variability of vulnerability indices^{34,63}, supporting its position at the lowest level of the hydrogeologic hierarchy. Assigning a higher weight would bias the assessment toward contamination-driven mapping, artificially inflating vulnerability in already contaminated locations while suppressing vulnerability in hydrogeologically sensitive zones that have not yet experienced degradation. LULC was mapped using GIS and included bare land (0.01%), artificial surfaces (0.33%), cultivated land (77.62%), grassland (12.09%), forest (9.83%), and water bodies (0.12%). Cultivated land dominates making agricultural contamination a major groundwater concern⁴¹. Thus, it can establish the amount of contaminated water that has been discharged into the aquifer and the amount of contamination that is being transported downstream.

Process-based framework for aquifer vulnerability assessment

This study advances AVA by explicitly bridging the long-standing gap between physically based hydrological processes and empirical VI indexing approaches. Traditional DRASTIC-type frameworks rely on static, often empirically inferred hydrogeological layers, which limits their ability to represent dynamic SGWI and evolving anthropogenic pressures. By integrating transient SWAT–MODFLOW simulations with enhanced DRASTIC variants, the proposed framework moves beyond purely intrinsic vulnerability mapping toward a more process-informed assessment. The incorporation of integrated and physical simulation improves the realism of key DRASTIC parameters, particularly depth to water table and net recharge, which are known to exert dominant control on contaminant transport in unconfined aquifers. Compared with conventional empirical parameterization, the SWAT–MODFLOW coupling reduces structural uncertainty and ensures internal consistency among hydrological variables. This represents a substantive methodological progression over earlier studies that applied DRASTIC in isolation or relied on lumped recharge estimates. DRASTIC and DRASTICL are powerful tools for AVA when optimized using a well-calibrated integrated SWAT–MODFLOW model. Once the corresponding rates and weights of the parameters were assigned, the AVA was determined using a dimensionless VI. The DRASTIC index spans 69–210, whereas the DRASTICL index extends from 84 to 225, indicating an upward shift in vulnerability intensity when land use effects are incorporated (Fig. 3). Despite broadly consistent spatial patterns (Fig. 3), both models were dominated by moderate vulnerability, covering 42.07% (DRASTIC) and 41.49% (DRASTICL) of the study area. However, a systematic redistribution is evident: the proportion of high- to very-high-vulnerability zones increases from 21.71% in DRASTIC to 24.13% in DRASTICL, whereas low-vulnerability classes slightly contract. This shift reflects the sensitivity of the DRASTICL framework to anthropogenic controls, suggesting that including land use factors amplifies vulnerability signals in already susceptible zones rather than altering the overall spatial hierarchy. Consequently, low-to-moderate vulnerability classes remain the most spatially extensive, whereas high-risk zones become more noticeable under DRASTICL, implying a stronger representation of pollution susceptibility in areas influenced by surface activities (Table 3).

Although only a small portion of the area is extremely vulnerable, these zones are located near important water supplies and industrial sites, indicating high intrinsic vulnerability. Therefore, it is of utmost importance to protect groundwater in these extremely vulnerable areas to ensure that pollutants do not seep into the water supply sources. The DRASTICL model shows a reduction in very high vulnerability zones and a redistribution toward high vulnerability classes, indicating a shift rather than uniform amplification. The most noticeable shift caused by the human impact of agricultural growth was in the net recharge rating. Incorporating optimised parameters into the AVA is expected to yield reliable findings that are more precise and reliable. It is anticipated that optimizing parameters during the calibration of a physically based integrated model will provide significant insights into determining the AVA. Through optimization of the parameters, a more precise description of these hydrogeological processes will be made possible, which will increase the reliability of the AVA. Parameter optimization provides more in-depth knowledge of the hydrogeological processes that influence groundwater flow and contaminant transport throughout the environment. This study provides a comprehensive method for AVA by combining DRASTIC/L and W methodologies. This study contributes to a more comprehensive and specific understanding of aquifer vulnerability to pollution.

Model	Class range	Rating	Areal coverage (%)
DRASTIC	<95	Very low	5.21
	95–125	Low	31.02
	125–153	Moderate	42.07
	153–180	High	20.55
	>180	Very high	1.16
DRASTICL	<110	Very low	4.47
	110–140	Low	29.49
	140–168	Moderate	41.49
	168–200	High	23.57
	>200	Very high	0.66
DRASTICLW	<118	Very low	3.87
	118–150	Low	30.66
	150–180	Moderate	39.31
	180–210	High	23.72
	>210	Very high	2.45

Table 3. Vulnerability class indices for the DRASTIC, DRASTICL, and DRASTICLW models.

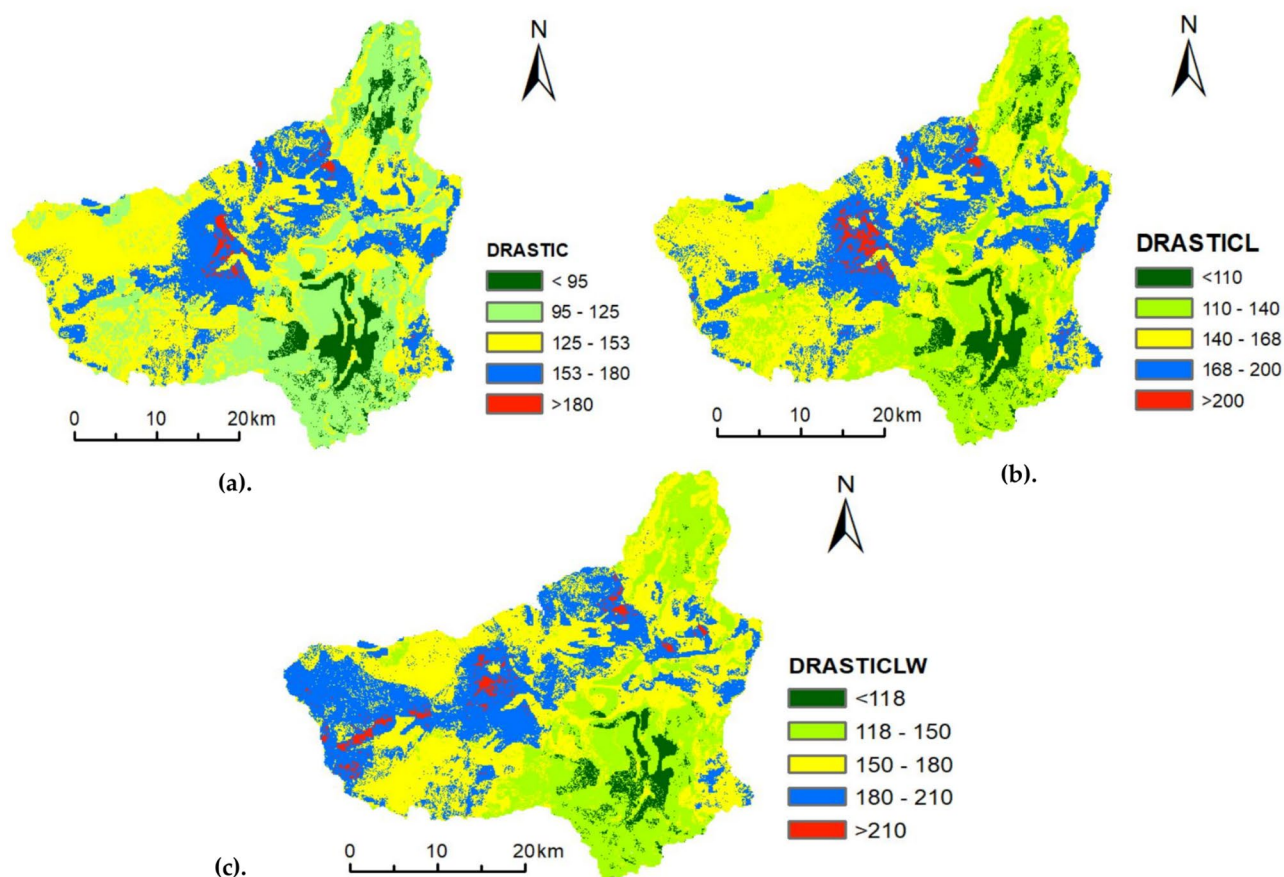


Fig. 3. Aquifer vulnerability distribution maps for an improved (a). DRASTIC, (b). DRASTICL, (c). DRASTICLW.

Bridging intrinsic aquifer vulnerability with WQI integration

This study bridges intrinsic aquifer vulnerability and water quality by embedding empirical contamination signals through the Water Quality Index (WQI) into a physically based vulnerability framework, enabling alignment between hydrogeological susceptibility and observed groundwater degradation patterns. (Fig. 2) illustrates the spatial distribution of key vulnerability parameters using perceptually uniform color scales, with warmer colors representing higher vulnerability or poorer water quality conditions and cooler colors indicating

lower vulnerability. The WQI values were spatially interpolated using ArcGIS, as shown in (Fig. 1). Color scales were selected to ensure intuitive interpretation of vulnerability gradients, with red hues consistently representing higher-risk or non-potable conditions, following common practice in groundwater vulnerability mapping. The location with the lowest level of pollution had a WQI score of 24.62. There was variation in the distribution of water quality throughout the region, overing 0.043%, 13.42%, 34.32%, 39.23%, and 12.99% for categories of excellent, good, bad, very poor, and unsuitable water quality, respectively. Some sites were characterized by excellent and good water quality and were situated in the western and northwestern segments. Regions classified as having poor and very poor water quality accounted for 34.32% and 39.23% of the area, respectively. Very high-water quality accounted for only 0.043% of the total area. By designating zones with varying degrees of water quality, decision-makers can effectively allocate resources and implement adaptable plans to enhance overall water quality. This offers thorough information on water quality dynamics to promote reliable decision-making for sustainable water management.

Modified DRASTICL approach that reflects the impacts of surface pollutant loading was used to assess individual susceptibility to contamination. A comparative analysis between the DRASTIC and DRASTICL index layers was conducted to evaluate differences in spatial vulnerability patterns. Validation of the AVA models was conducted through comparative evaluation of their ability to reproduce spatial patterns of groundwater contamination. Because WQI is incorporated as an input to the DRASTICLW model, it is not used as an independent validation dataset. Instead, performance is assessed by comparing the relative spatial consistency and discrimination of the different vulnerability frameworks. High vulnerability indices corresponded to regions exposed to contamination, as confirmed by the validation of DRASTIC and DRASTICL vulnerability maps. The findings from the DRASTICL approach were closely aligned with those from the DRASTIC method. However, the latter classified some locations as moderately susceptible, indicating fewer areas of high vulnerability. Most of the area exhibits moderate vulnerability, primarily due to cultivated land that requires high protection. Human activities that discharge contaminants to the surface contribute significantly to groundwater pollution, as these pollutants infiltrate the aquifers.

Through water quality analysis, this study identified agricultural activity as the main source of groundwater pollution, indicating contamination stemming from excessive agricultural practices. The moderately vulnerable areas exhibit higher WQI concentrations. Hence, enhancing the accuracy of groundwater risk assessment models is crucial for monitoring human activities to achieve groundwater sustainability. Both DRASTIC and DRASTICL capture similar spatial vulnerability structures, indicating stable baseline representation across model formulations. This was achieved by improving the net recharge factor, which is the primary input for both the DRASTIC and modified DRASTICL methods. The effectiveness of AVA can be strengthened by utilizing the DRASTIC and DRASTICL models along with the integrated SWAT-MODFLOW model. This underscores the importance of obtaining precise and comprehensive data to evaluate aquifer vulnerability. This finding underscores that selecting an appropriate AVA approach is pivotal for ensuring long-term and sustainable groundwater management.

Although the DRASTIC and DRASTICL models have been extensively employed in AVA applications, their relatively weak correlations with specific water quality parameters often result in the underestimation or overestimation of actual groundwater pollution risk. Several studies have reported that the correlation coefficients between nitrate concentrations and DRASTIC-based VI are typically low, indicating limited predictive reliability^{64,65}. Even when land use information is incorporated, as in DRASTICL, the improvement in correlation with water quality indicators remains marginal because the model structure does not explicitly account for contaminant loading dynamics or SGWI processes. This structural limitation suggests that conventional index-based methods primarily capture intrinsic susceptibility but fail to represent the empirical demonstration of contamination driven by anthropogenic stressors. To address this methodological gap, the present study advanced the framework supplementing the traditional DRASTIC parameters with an additional land use factor and normalized WQI to create integrated DRASTICLW. The introduction of an external term (WQI) tends to shift vulnerability values uniformly, potentially reducing the divergence among hydrogeologically distinct areas. This enhancement enables the framework to couple intrinsic vulnerability with actual water quality outcomes, thereby improving its responsiveness to contaminant-specific risks and yielding a more realistic appraisal of aquifer vulnerability under complex SGWI conditions. These decisions preserve the intrinsic nature of aquifer vulnerability assessment while enabling adaptive refinement based on observed water quality conditions.

Robustness and generalization of DRASTICLW framework

The DRASTICLW framework represents a substantive advancement over conventional DRASTIC and DRASTICL formulations by embedding normalized water quality information (WQI_{norm}) as an adaptive weighting layer that links intrinsic hydrogeologic susceptibility with observed contamination patterns. This integration enables a contamination-responsive vulnerability, as reflected in the spatial distribution of AVA (Fig. 3c), which exhibits enhanced normalization relative to the baseline DRASTIC models (Fig. 3a–b). The resulting gradient from low to high index values delineates the full vulnerability spectrum, where elevated DRASTICLW values correspond to areas where hydrogeologic susceptibility coincides with degraded water quality, and lower values indicate relatively resilient aquifer conditions. The framework occupies an intermediate position between purely intrinsic vulnerability mapping and contaminant-specific risk assessment, offering a pragmatic solution for regional groundwater management under data-limited conditions. By incorporating WQI_{norm} as a dynamic constraint, DRASTICLW introduces an adaptive feedback mechanism that enhances model discrimination, reduces predictive bias, and mitigates both underestimation of contaminated zones and overestimation in chemically stable regions. Model outputs (Table 3) demonstrate substantial spatial variability, with index values ranging from 88 to 242 (mean = 163.8; SD = 25.5). Vulnerability classes were defined as very low (88–118), low (118–150), moderate (150–180), high (180–210), and very high (>210), covering 3.87%,

30.66%, 39.31%, 23.72%, and 2.45% of the study area, respectively. High-vulnerability zones are predominantly associated with intensive agricultural and industrial regions, whereas lower values correspond to less disturbed hydrogeologic settings.

The integration of WQI enhances hydrogeochemical coherence by incorporating a composite proxy of cumulative contaminant loading. WQI aggregates ionic and nutrient parameters into a dimensionless indicator of aquifer degradation⁴⁷, and has been shown to correlate strongly with key tracers such as nitrate and total dissolved solids^{33,66}. High WQI values commonly co-occur with shallow groundwater levels, permeable vadose zones, and high recharge conditions—factors inherently associated with high vulnerability in DRASTIC-based frameworks^{14,34}. This alignment reinforces the consistency of the integrated SWAT–MODFLOW–DRASTICLW approach, where WQI serves not merely as a concentration metric but as an empirical constraint reflecting underlying transport processes. A key strength of DRASTICLW lies in its ability to balance model specificity and generalization across heterogeneous hydrogeologic settings. Traditional DRASTIC calibrations based on single contaminants are prone to overfitting localized anomalies or underrepresenting broader contamination patterns³³. In contrast, the use of WQI introduces a multi-parameter aggregation effect that captures cumulative contamination signals while reducing local stochastic noise⁶⁷. This implicit regularization enhances spatial stability, improves discrimination, and supports robust performance under conditions of data sparsity and scale heterogeneity^{13,68}.

The inclusion of WQI as a normalized constraint further strengthens the framework by linking intrinsic vulnerability with observed contamination patterns. As a result, DRASTICLW provides more spatially coherent and management-relevant vulnerability maps, particularly in complex basins where interactions among recharge variability, groundwater depth, and land use pressures govern contamination risk. The framework is transferable to other regions where hydrological model outputs and groundwater quality data are available. Its structure—combining process-based simulation with normalized empirical constraints—offers flexibility across diverse hydrogeological and climatic settings without altering the core methodology. Overall, DRASTICLW represents a hybrid, process-informed, and data-constrained advancement in aquifer vulnerability assessment, supporting improved diagnostic reliability, spatial prioritization, and sustainable groundwater management. While individual pollutants exhibit distinct mobility and attenuation characteristics, the adopted framework captures cumulative susceptibility relevant for regional groundwater protection planning.

Parameter sensitivity analysis

Sensitivity analysis (SA) was performed using a map-removal approach to quantify the relative influence of each parameter on the vulnerability indices. Parameter importance was evaluated based on deviations in the mean index and changes in variability (standard deviation, SD) following the sequential exclusion of the individual layers. The results (Table 4) show that aquifer media and the vadose zone exert the strongest influence, as their removal produces the largest increases in SD, indicating a high spatial dependence of vulnerability on subsurface properties controlling contaminant transport and attenuation. Recharge and groundwater depth exhibited moderate to high sensitivity, consistent with their roles in governing contaminant flux and travel time. In contrast, topography produced minimal changes in both the mean and variability, indicating a limited contribution to subsurface contamination processes. The relatively stable response observed for groundwater depth reflects a consistent spatial influence rather than dominance. The inclusion of the WQI in the DRASTICLW framework preserves the sensitivity hierarchy while enhancing spatial differentiation by incorporating empirical contamination signals. Despite localized variations, the overall spatial structure of vulnerability remained stable across parameter removal scenarios, indicating the structural robustness of the model. Overall, parameters that directly control subsurface flow and transport processes dominate the vulnerability patterns, whereas surface-related parameters exert a secondary influence.

Progressive contamination hotspots

The progressive increase in the composite vulnerability index from DRASTIC to DRASTICL and ultimately to DRASTICLW follows a logical methodological hierarchy that reflects the successive enrichment of information content within the model. The original DRASTIC model¹⁷ quantifies intrinsic vulnerability, emphasizing the physical capacity of an aquifer to transmit contaminants based solely on hydrogeologic parameters such as depth to water, recharge, and aquifer media. This baseline representation assumes uniform exposure risk, neglecting the spatial heterogeneity of human pressures. The modified DRASTICL model introduces a land use weighting layer, capturing the influence of surface activities such as agriculture, urbanization, and industrial zones on potential contaminant loading pathways^{30,35}. This adjustment increases the composite index because anthropogenic land cover amplifies the likelihood of pollutant infiltration through vulnerable hydrogeologic settings. DRASTICLW integrates the WQI as an empirical weighting factor that directly represents the realized contamination footprint derived from measured groundwater quality parameters^{57,69}. The inclusion of WQI elevates the index further by embedding multi-pollutant information reflecting both natural and human-induced degradation processes, thus translating potential vulnerability into actual contamination risk. The WQI surface reflects observed groundwater quality but does not account for the hydrogeological controls governing contaminant transport. In contrast, the DRASTICLW framework integrates hydrogeological susceptibility with empirical contamination patterns, thereby enabling improved spatial discrimination of areas that are both vulnerable and affected. Consequently, the systematic increase from DRASTIC to DRASTICL to DRASTICLW is not arbitrary but conceptually coherent: each iteration progressively enhances the explanatory power of the model by incorporating additional dimensions of contamination drivers first by anthropogenic exposure, then empirical hydrochemical evidence. The final DRASTICLW framework therefore provides a comprehensive, data-driven depiction of aquifer susceptibility that unites intrinsic sensitivity, land-surface influence, and observed water quality deterioration within a single, integrated VI.

Models	Removed parameter	Min.	Max.	Mean	SD
DRASTIC	D	62	190	122	21
	R	65	182	115	17
	A	63	195	124	22
	S	63	175	110	20
	T	51	180	110	21
	I	57	184	121	22
	C	57	183	116	21
DRASTICL	D	62	190	122	21
	R	80	205	132	18
	A	78	211	141	22
	S	74	192	127	20
	T	66	195	126	22
	I	72	210	138	23
	C	72	198	132	22
	L	69	210	133	23
DRASTICLW	D	64	203	132.4	23.6
	R	81	214	142.1	19.0
	A	79	227	151.4	25.0
	S	80	213	137.6	22.8
	T	67	209	136.6	24.5
	I	76	232	151.0	25.5
	C	74	221	142.6	24.0
	L	70	213	143.7	24.9
	W	84	225	150.3	23.1

Table 4. Sensitivity rates of the of DRASTIC, DRASTICL and DRASTICLW factors.

Threshold validation for adaptive aquifer management

The DRASTIC, DRASTICL, and DRASTICLW approaches assign weights to hydrogeologic parameters according to their relative importance in predicting contamination risk. Although DRASTIC, DRASTICL, and DRASTICLW produce indices with different numerical ranges, model comparison was conducted based on relative spatial patterns and validation performance rather than absolute index values. All vulnerability outputs were normalized prior to validation to ensure comparability across model formulations. The improved AVA generated in this study yields more realistic vulnerability maps, delineating zones with differing levels of susceptibility. Using integrated modeling (SWAT–MODFLOW), we demonstrated how parameter optimization and thematic enhancement strengthen AVA outcomes. The ROC analysis (Fig. 4) demonstrates that the original DRASTIC method achieves an area under the curve (AUC) of 0.86, indicating that intrinsic hydrogeologic characteristics alone explain a large portion of the spatial variability in groundwater risk. However, despite parameter refinement, DRASTIC remains structurally limited because it excludes anthropogenic stressors, which are dominant drivers of contemporary groundwater contamination. The modified DRASTICL approach partially addresses this gap by incorporating land use as an additional thematic layer. Land use acts as a proxy for human-driven contaminant loading (e.g., fertilizer application, industrial discharge, urban runoff), and the discrimination performance improves accordingly (AUC = 0.89), confirming that explicit representation of anthropogenic pressure enhances the model’s capacity to depict *specific* vulnerability rather than intrinsic potential. To ensure that vulnerability classes reflect statistical behavior rather than subjective decisions, aquifer vulnerability was classified using the mean and standard deviation of the DRASTIC-derived index. Because DRASTIC-based indices integrate heterogeneous drivers exhibit a continuous statistical distribution rather than discrete groupings³⁰.

The mean represents the baseline vulnerability condition, while the standard deviation quantifies deviations driven by hydrogeologic heterogeneity and contaminant loading intensity. Applying the mean–standard deviation framework allows statistically defensible separation of naturally protected zones from extreme hotspots, which typically coincide with shallow water tables, permeable vadose zones, and strong SGWI. This approach minimizes misclassification, particularly the masking of localized contamination areas, because hotspots are detected based on statistically significant deviation rather than arbitrary visual categorization. Classification via mean–standard deviation enables reproducibility and comparability across temporal simulations or management scenarios, a critical requirement when evaluating the effects of environmental variability. Within the DRASTICLW framework, this statistical classification improves hotspot detection by aligning intrinsic vulnerability patterns with empirical contamination signals. The proposed framework occupies an intermediate position between purely intrinsic vulnerability mapping and fully contaminant-specific risk assessment, offering a practical compromise for regional groundwater management under data constraints. The DRASTICLW framework represents a measurable methodological advancement over the DRASTIC and DRASTICL formulations by

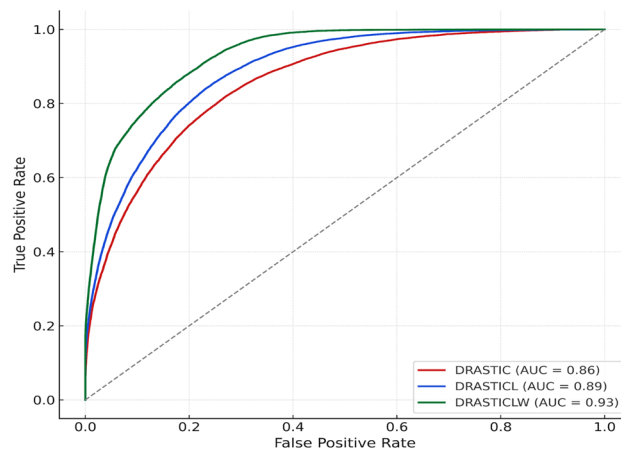


Fig. 4. Performance of DRASTIC, DRASTICL, and DRASTICLW vulnerability indices.

embedding direct water quality evidence (WQI) into the intrinsic vulnerability structure. While DRASTIC provides a baseline and DRASTICL introduces anthropogenic influence via land use weighting, DRASTICLW couples intrinsic susceptibility with measured contamination intensity, producing an empirically grounded representation of groundwater risk. Its feedback-responsive weighting dynamically adjusts vulnerability scores according to WQI magnitudes: aquifers with moderate intrinsic vulnerability but degraded water quality are up-weighted, revealing previously obscured hotspots, whereas zones with high intrinsic vulnerability but good water quality are down-weighted, reducing false positives. This bidirectional modulation improves predictive realism, as evidenced by the highest ROC performance (AUC = 0.93) (Fig. 4). The improved predictive performance of the DRASTICLW model, as reflected by higher AUC values relative to DRASTIC and DRASTICL, confirms that coupling intrinsic vulnerability with contamination-informed weighting enhances the model's ability to discriminate between impacted and less impacted areas. This finding aligns with growing evidence that hybrid approaches—combining process understanding with observational constraints—outperform purely intrinsic or purely empirical methods when applied at regional scales^{29,70}. However, uncertainty in AVA arises from multiple sources, including data availability, model structure, parameterization, and spatial classification. The integration of physically based hydrological processes improves the delineation of vulnerability zones. It also enhances spatial consistency, particularly in areas influenced by anthropogenic activities. The statistical classification of vulnerability indices using a mean–standard deviation approach provides a reproducible basis for comparative analysis, though it necessarily simplifies spatially autocorrelated patterns. Future work could further refine classification schemes by incorporating spatial statistics or probabilistic clustering techniques.

In particular, sparse monitoring networks may fail to capture localized contamination hotspots, whereas temporal averaging of water quality observations can mask short-term variability associated with seasonal recharge and land use dynamics. Collectively, these factors constrain the precision of vulnerability estimates, especially in data-limited regions. By explicitly incorporating hydrogeologic conditions, land use stressors, and empirical monitoring data, DRASTICLW captures the dynamic interplay between aquifer characteristics and contamination pressures. The transition from DRASTIC → DRASTICL → DRASTICLW therefore reflects more than a methodological refinement: it enhances the credibility and decision value of vulnerability maps, enabling groundwater managers to identify contamination hotspots, optimize monitoring networks, and prioritize mitigation interventions. WQI-only mapping exhibited lower discriminatory performance than the integrated framework, demonstrating added predictive value beyond empirical observation alone. DRASTICLW demonstrates a statistically measurable improvement in discriminatory capability relative to conventional DRASTIC-based models. Although vulnerability indices exhibit spatial autocorrelation, the adopted classification scheme provides a consistent basis for comparative model evaluation at the basin scale. The vulnerability maps provide a spatial basis for prioritizing groundwater protection zones, guiding monitoring network design, and supporting land use planning in areas where shallow aquifers and intensive agriculture coincide. The proposed framework is transferable to other basins where basic hydroclimatic, land use, and groundwater quality data are available, with recalibration of model components required to reflect local hydrogeological conditions. The modular structure of the framework facilitates adaptation to contrasting environmental settings without altering its conceptual foundation. Ultimately, these findings demonstrate that integrating intrinsic vulnerability with observed water quality signals yields actionable intelligence for adaptive aquifer management in regions experiencing escalating anthropogenic and climatic stresses.

Implications for sustainable groundwater management

The significance of regional aquifer vulnerability assessment lies not in mapping alone, but in its ability to translate hydrogeological process understanding into spatial decision support. By coupling physically based surface–groundwater simulations with empirically informed VI, the proposed framework supports management interpretation in hydrodynamic behavior rather than heuristic classification. Highly vulnerable zones correspond to hydrogeological conditions that favor rapid contaminant transmission configurations that promote surface

loading. These areas represent structurally sensitive portions of the aquifer where contaminant residence times are short and attenuation capacity is limited. From a management perspective, they constitute priority targets for preventative protection. Vulnerability gradients enable spatial prioritization of nutrient controls, diffuse pollution mitigation, and recharge-area protection, aligning interventions with the physical architecture of the aquifer rather than applying uniform measures across heterogeneous settings. The framework also supports rational groundwater monitoring design. Conventional networks often reflect administrative constraints more than hydrogeological sensitivity. By identifying recharge-dominated zones and shallow flow systems, the vulnerability maps guide monitoring toward locations where early contamination signals are most likely to emerge. This shifts practice from retrospective assessment toward anticipatory surveillance, strengthening early detection of aquifer degradation.

Beyond monitoring, the process-informed structure enhances land and resource planning. Vulnerability patterns reveal where surface activities are strongly coupled to subsurface response, clarifying areas where land use intensification may impose disproportionate pressure on groundwater systems. In rapidly developing basins, such information informs infrastructure siting, agricultural expansion, and waste management decisions while respecting hydrogeological constraints supporting balanced trade-offs between development and aquifer protection. The framework advances preventive groundwater management by emphasizing susceptibility rather than observed contamination alone. While water quality measurements describe current system state, vulnerability analysis anticipates future response under continued pressure. Integrating these perspectives enables differentiation between transient anomalies and structurally persistent risk zones. In data-limited settings, this forward-looking prioritization provides a crucial basis for enhanced groundwater governance. Although vulnerability maps reveal the spatial organization of hydrogeological susceptibility. When combined with regulatory oversight, targeted hydrochemical investigation, and stakeholder engagement, the framework offers a scalable foundation for sustainable groundwater management. Hence, AVA evolves from descriptive mapping to operational hydrogeological intelligence. The spatial structure of the vulnerability patterns reflects the hydrogeological architecture controlling contaminant transmission. The alignment of high vulnerability with shallow groundwater, recharge concentration zones, and intensive land use demonstrates hydraulic coupling between surface pressures and subsurface response. This represents spatial expressions of flow connectivity, residence time, and attenuation capacity, enabling management decisions that anticipate system behavior.

Despite its advances, the integration of WQI relies on available groundwater quality observations, which may be temporally sparse and unevenly distributed. All regional vulnerability assessments involve uncertainty arising from data limitations, interpolation, model structure, and temporal aggregation. While physically based simulations reduce structural inconsistencies, residual uncertainty remains inherent to basin-scale representation. Monitoring feedback and updated datasets refine system understanding over time, allowing agencies to prioritize protection while retaining strategic flexibility. This adaptive perspective positions AVA as a dynamic decision-support tool aligned with evolving hydrogeological conditions. In addition, both hydrological and groundwater quality data are temporally aggregated, potentially limiting the representation of short-term extremes associated with seasonal recharge events or episodic contamination. The framework evaluates generalized vulnerability rather than pollutant-specific risk, as it is based on aggregated water quality indicators rather than individual contaminants. While this approach may mask distinct transport and attenuation behaviors of specific pollutants, groundwater quality indicators represent contamination outcomes rather than explicit pressure metrics. In data-limited regions, however, they provide an integrated and empirically defensible proxy for cumulative anthropogenic influence. Future research should therefore explore pollutant-specific extensions of the framework, incorporate explicit anthropogenic pressure datasets, and assess uncertainty propagation through ensemble-based hydrological simulations to further strengthen the framework. Application of the approach across multiple basins with contrasting hydrogeological and climatic settings would further strengthen its general applicability and enhanced explanatory coherence for groundwater management under increasing anthropogenic stress.

Conclusion

This study demonstrates a process-informed advancement in aquifer vulnerability assessment by integrating physically-based surface–subsurface interaction modeling with modified DRASTIC indices and empirical groundwater quality constraints. By coupling SWAT–MODFLOW simulations with DRASTIC-based mapping, the framework overcomes key limitations of conventional index approaches that rely on static or empirically inferred parameters. The resulting vulnerability maps more realistically capture the combined effects of recharge dynamics, groundwater depth, and land use influences on contaminant susceptibility. Incorporation of a normalized WQI_{norm} as an empirical constraint further enhances model performance by aligning intrinsic vulnerability patterns with observed groundwater quality while preserving hydrogeological consistency. Its modular structure and reliance on increasingly accessible datasets support application across diverse hydrogeological settings, including data-limited regions. While the approach introduces additional data and computational requirements, these are justified by improved reliability, and the ability to delineate contamination hotspots more accurately. This enhanced capability supports targeted and efficient groundwater management, especially in complex hydrogeological conditions and anthropogenic pressures. Future applications in diverse hydroclimatic and hydrogeological settings will further evaluate the generalizability and operational benefits of the framework. Overall, the framework offers a scalable, transferable, and practically applicable methodology for advancing groundwater protection and sustainable resource management.

Data availability

The datasets used during the current study available from the corresponding author on reasonable request.

Received: 9 December 2025; Accepted: 27 April 2026

Published online: 11 May 2026

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Author contributions

T.D.M.: Conceptualization, Methodology, Writing original draft, Reviewing and Editing, Visualization and Validation; I.-M.C.: Resources, Reviewing and Editing, Validation, Funding acquisition; S.W.C.: Visualization and Validation, Reviewing and Editing.

Funding

This Research was carried out under the KICT Research Program (Grant No. 20260156-001) funded by the

Ministry of Science and ICT, South Korea.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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